STOCHASTIC INVENTORY MODELING

Notation

c: unit purchase cost ($/unit)
b: unit loss-of-goodwill cost ($/unit)
v: unit salvage revenue ($/unit)
K: fixed ordering cost ($/order)
i: annual inventory carrying rate ($/$/yr)
h=ic: unit annual inventory holding cost ($/unit/yr)
λ: demand rate (units/yr)
q: order quantity (units)
r: reorder level (units)
$n(r)$: expected number of shortages in a cycle when the reorder level is r (units)
s: expected safety stock level (units)
X_L: demand during lead time, a random variable with pdf $f(x_L)$; cdf $F(x_L)$; $E[X_L]$ =; and $Var(X_L) = \sigma_L^2$
μ_D: expected demand
σ_D^2: variance of demand
μ_τ: expected lead time
σ_τ^2: variance of lead time

Leibnitz's Rule

$$\frac{d}{dy}\int_{\ell(y)}^{u(y)} h(x,y)dx = \int_{\ell(y)}^{u(y)} \frac{\partial h(x,y)}{\partial y}dx + h(u(y),y)u'(y) - h(\ell(y),y)\ell'(y)$$

Periodic Review Systems

$$F(q^*) = \frac{c_u}{c_u + c_o}$$

	Underage	Overage
Single-period: Cost min.	c_u	c_o
Single-period: Profit max.	$p - c + b$	$c - v$
Multiple-period: Full back.	b	h
Multiple-period: Lost sales	$p - c + b$	h

Demand during Lead Time (DDLT) Parameters

$\mu_L = \mu_D \mu_\tau$
$\sigma_L^2 = \mu_\tau \sigma_D^2 + \mu_D^2 \sigma_\tau^2$

Continuous Review Systems

$$n(r) = \int_r^\infty (x_L - r)\, f(x_L) dx_L$$

If the demand is normally distributed, we have

$$n(r) = \sigma_L L(z) = \sigma_L L\left(\frac{r - \mu_L}{\sigma_L}\right)$$

Full Backlogging

$$s = r - E[X_L]$$

$$q^* = \sqrt{\frac{2\lambda[K + bn(r)]}{h}}$$

$$1 - F(r^*) = \frac{qh}{b\lambda}$$

$$C(q,r) = c\lambda + \frac{K\lambda}{q} + h\left(\frac{q}{2} + s\right) + \frac{bn(r)\lambda}{q}$$

Lost Sales

$$s = r - E[X_L] + n(r)$$

$$q^* = \sqrt{\frac{2\lambda[K + bn(r)]}{h}}$$

$$1 - F(r^*) = \frac{qh}{qh + b\lambda}$$

$$C(q,r) = c\lambda + \frac{K\lambda}{q} + h\left(\frac{q}{2} + s\right) + \frac{bn(r)\lambda}{q}$$

Type I Service Level

$$q = \sqrt{\frac{2K\lambda}{ic}}$$

$$F(r) = \alpha$$

Type II Service Level

$$q = \sqrt{\frac{2K\lambda}{ic}}$$

$$n(r) = (1 - \beta)q$$

LOGISTICS

Notation
h: in-transit inventory holding cost ($/unit/year)
λ*:* demand rate (units/yr)
τ*:* lead time (years)

In-transit inventory holding cost
$h\lambda\tau$

FORECASTING

Notation
T: planning horizon length indexed by t
D_t*:* observed demand in period t
F_t*:* demand forecast for period t
E_t*:* forecast error in period t
S_t*:* value of the intercept at the end of period t
G_t*:* value of the slope at the end of period t
π*:* forecasting period

Forecast Error Term
$E_t = F_t - D_t$

Forecasting Error: Mean Absolute Deviation

$$MAD_n = \frac{1}{n}\sum_{t=1}^{n} |E_t|$$

Forecasting Error: Mean Squared Error

$$MSE_n = \frac{1}{n}\sum_{t=1}^{n} E_t^2$$

Forecasting Error: Mean Absolute Percentage Error

$$MAPE_n = \frac{1}{n}\sum_{t=1}^{n} \left|\frac{E_t}{D_t}\right| 100$$

Stationary: Moving Average

$$F_t = \frac{1}{N}\sum_{i=t-N}^{t-1} D_i$$

Stationary: Simple Exponential Smoothing
$F_t = \alpha D_{t-1} + (1-\alpha)F_{t-1}$

Linear Trend: Double Exponential Smoothing

$$S_t = \alpha D_t + (1-\alpha)(S_{t-1} + G_{t-1})$$

$$G_t = \beta(S_t - S_{t-1}) + (1-\beta)G_{t-1}$$

$$F_{t,t+\pi} = S_t + \pi G_t$$

Linear Trend: Linear Regression Analysis

$$\hat{b} = \frac{\sum_{t=1}^{T}(t-\bar{t})(D_t - \bar{D})}{\sum_{t=1}^{T}(t-\bar{t})^2} = \frac{T\sum_{t=1}^{T} D_t t - (\sum_{t=1}^{T} D_t)(\sum_{t=1}^{T} t)}{T\sum_{t=1}^{T} t^2 - (\sum_{t=1}^{T} t)^2}$$

$$\hat{a} = \bar{D} - \hat{b}\bar{t}$$

$$F_{T,T+\pi} = \hat{a} + (T+\pi)\hat{b}$$

$$\bar{D} = \frac{\sum_{t=1}^{T} D_t}{T}$$

$$\bar{T} = \frac{\sum_{t=1}^{T} t}{T}$$

$$\sum_{t=1}^{T} t = \frac{T(T+1)}{2}$$

$$\sum_{t=1}^{T} t^2 = \frac{T(T+1)(2T+1)}{6}$$